NYC Citibike Demand Analysis: Member vs Casual Ridership

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Abstract

On a working day in New York City, Citibike gains 461 member rides and loses 520 casual rides compared to a weekend, a net divergence of nearly 1,000 trips per day that is invisible in aggregate demand. We uncover this by fitting separate Bayesian Structural Time Series (BSTS) models to member and casual daily counts across three years of data (2022–2024, $n = 1{,}065$). Temperature affects both groups similarly (+29–40 rides/°C), and season effects vanish once temperature is controlled for. The implication: member and casual ridership are distinct products that require separate operational and promotional strategies.

1 Introduction

Citibike members pay a recurring subscription and primarily commute; casual riders pay per trip and ride for leisure or tourism. If these groups respond differently to the same conditions, aggregate demand models obscure actionable patterns. This report tests that hypothesis using daily Citibike trip data (January 2022 to December 2024) merged with weather observations, fitting separate BSTS models for each rider type, comparing against OLS baselines, and translating the findings into operational recommendations. Code and data pipeline: GitHub Repository.

2 Data

2.1 Dataset Overview

The dataset contains 1,065 daily observations spanning January 2022 to December 2024 (July 2022 is absent due to missing source data). Each observation includes three ridership outcomes (`total_rides`, `member_count`, `casual_count`), nine weather variables (temperature, precipitation, wind, humidity, weather code), and six calendar variables (year, month, season, weekday, workingday, holiday).

2.2 Data Processing

Ingestion. Citibike trip records were downloaded from the Citibike system data portal and aggregated into daily counts by user type. Weather data were retrieved from the Open-Meteo API for the same period. Both datasets were merged by date and stored as partitioned CSV files on AWS S3, organized by year and month.

Preprocessing. WMO weather codes were grouped into six categories (Clear, Cloudy, Light Rain, Rain, Snow, Storm). Temperature variables were highly correlated ($r > 0.94$); only `temp_max` was retained. A VIF check confirmed no remaining multicollinearity (all $\mathrm{GVIF}^{1/(2\cdot\mathrm{Df})} < 2$).

3 Exploratory Analysis

Daily total rides ranged from 111 to 4,531 (mean 2,656), with a bimodal distribution driven by season: winter clusters around 500–2,000, summer near 3,000–4,000 (Figure 1). The most striking pattern is in Figure 2: member counts jump on workdays while casual counts drop. On weekends, casual ridership nearly matches members.

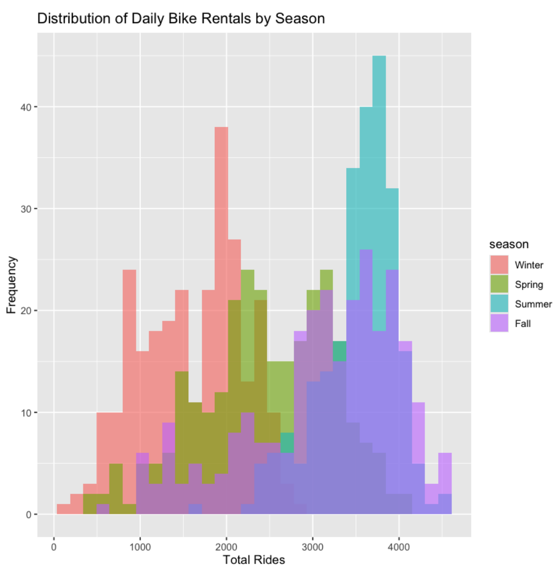

Figure 1: Distribution of daily total rides by season.

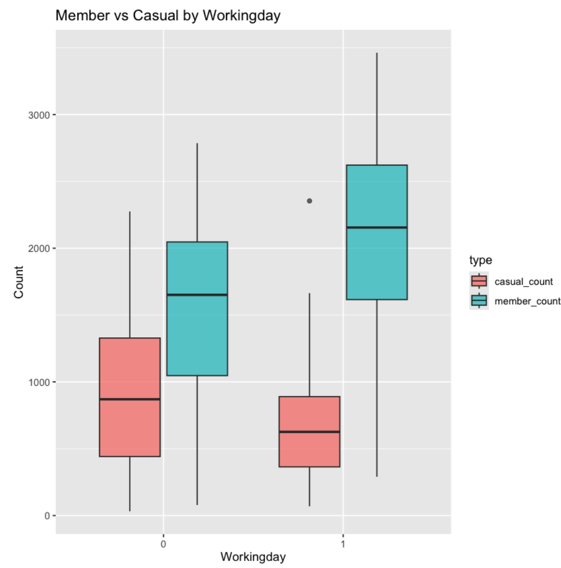

Figure 2: Member vs casual ridership by workingday status.

Temperature shows a positive association with both groups, but the slope is steeper for casual riders, suggesting greater weather sensitivity among leisure users (Figure 3). Precipitation shows a negative effect for both groups once extreme values are excluded (Figure 4), though the effect is concentrated at low levels.

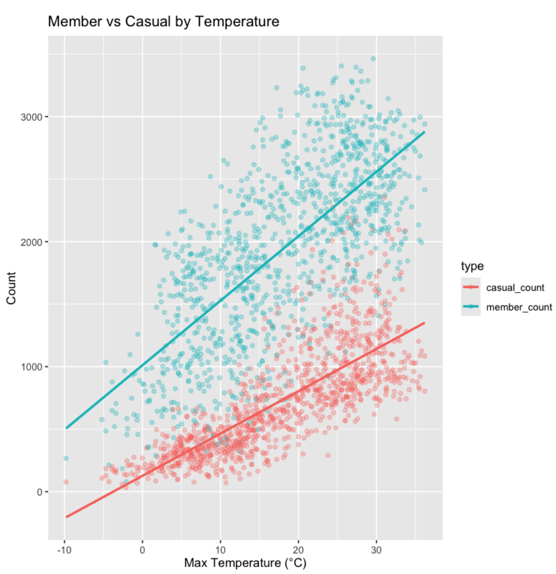

Figure 3: Ridership by maximum daily temperature. Lines are OLS fits.

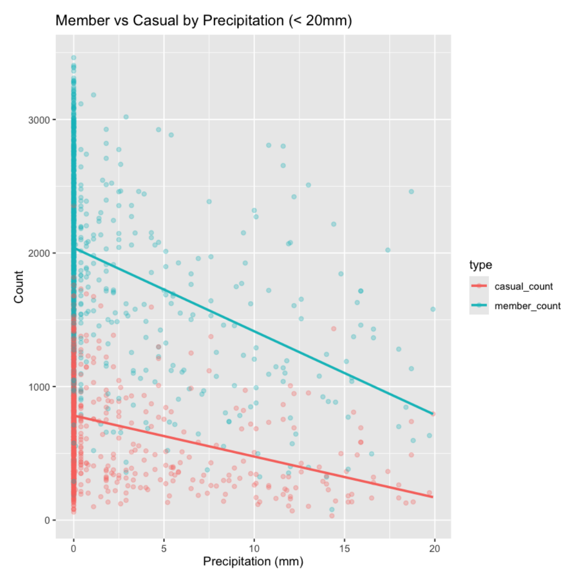

Figure 4: Ridership by precipitation (< 20mm). Lines are OLS fits.

A correlation analysis (Figure 5) revealed high collinearity among the three temperature variables (`temp_max`, `temp_min`, `atemp_mean`; $r > 0.94$), confirming that retaining only `temp_max` loses minimal information. Member and casual counts correlate strongly with each other ($r = 0.55$) and with temperature ($r \approx$ 0.7–0.8), but show near-zero correlation with precipitation and wind speed, suggesting that adverse weather suppresses demand through discrete events (rain, snow) rather than through continuous variation.

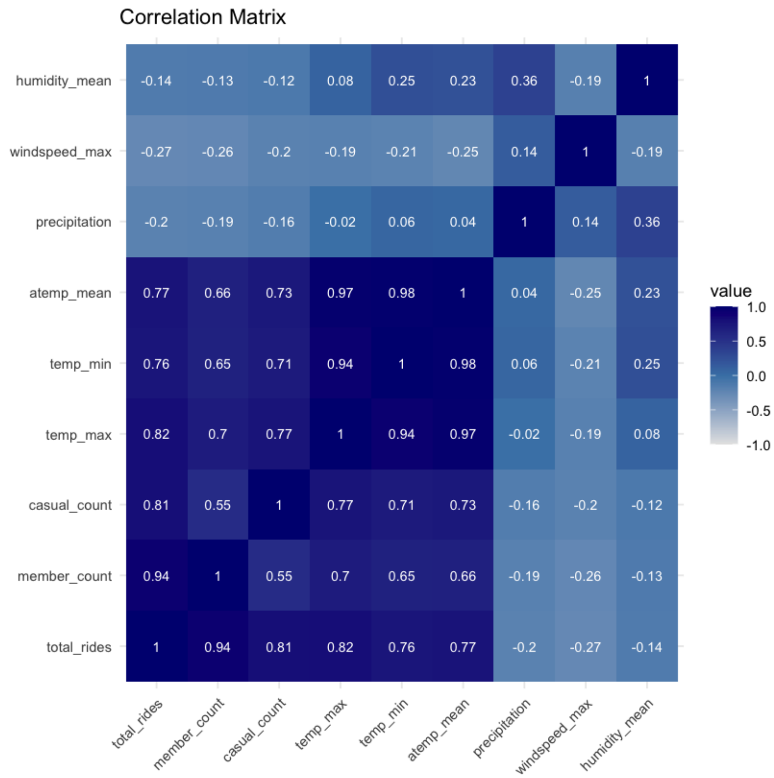

Figure 5: Correlation matrix of ridership and weather variables. Temperature variables cluster tightly ($r > 0.94$); precipitation and wind show weak linear association with ridership.

4 Methodology

We fit BSTS models [1] separately for member and casual counts using R's `bsts` package (1,000 MCMC iterations). Each model includes a local linear trend and weekly seasonal component ($s = 7$), with spike-and-slab variable selection assigning each predictor an inclusion probability (`inc.prob`). Predictors with `inc.prob` > 0.9 are considered strongly supported.
Two specifications per outcome yield four models:

Model	Specification
Baseline	weather + calendar covariates
Interaction	Baseline + `temp_max:workingday` + `precipitation:workingday`

OLS regression with the baseline specification serves as a benchmark for coefficient comparison.

5 Results

5.1 Coefficient Comparison Across Models

Figure 6 displays posterior coefficients with `inc.prob` > 0.9 across all four BSTS models, with `workingday1` showing the most prominent asymmetry between member and casual riders.

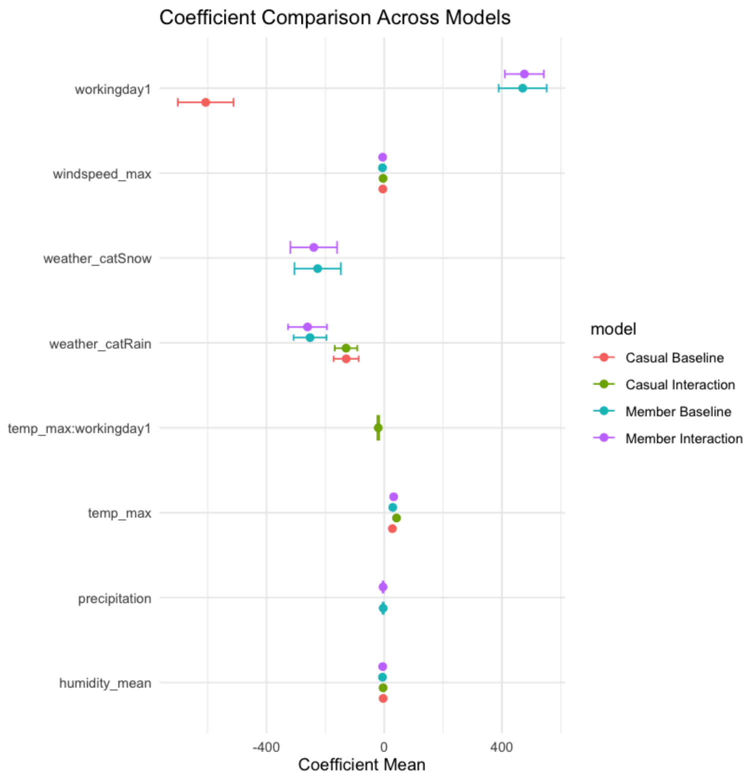

Figure 6: Posterior coefficients with `inc.prob` > 0.9. Points = posterior mean; bars = 95% credible interval.

5.2 Coefficient Summary

Table 1 reports posterior coefficients and inclusion probabilities across all four models. The dominant effect is `workingday1`: working days add +470 member rides but remove −607 casual rides in the Baseline, a net divergence of over 1,000 rides per day.

	Member				Casual			
	Baseline		Interaction		Baseline		Interaction	
Variable	Mean	Prob	Mean	Prob	Mean	Prob	Mean	Prob
`workingday1`	+470	1.00	+476	1.00	−607	1.00	−22	0.21
`temp_max`	+29	1.00	+32	1.00	+28	1.00	+42	1.00
`temp_max:workingday1`	—	—	+0	0.01	—	—	−20	1.00
`weather_catRain`	−252	1.00	−261	1.00	−129	1.00	−130	1.00
`weather_catSnow`	−226	1.00	−240	1.00	−0	0.00	−0	0.00
`precipitation`	−4	0.98	−4	0.98	−0	0.01	−0	0.00
`season`	—	≈0	—	≈0	—	≈0	—	≈0

Table 1: Posterior coefficients (Mean) and inclusion probabilities (Prob) across all four BSTS models. Means are rounded.

For members, neither interaction term is selected and `workingday1` remains stable at +476, reflecting commuting demand that is independent of weather. For casual riders, `temp_max:workingday1` (−20, Prob = 1.00) reveals that temperature sensitivity is concentrated on non-working days: on workdays, leisure demand is suppressed regardless of conditions, so temperature loses its predictive power. As shown in Table 2, R^2 improves negligibly with interaction terms, confirming the Baseline already captures the main effects.

5.3 Weather Effects and Season Absorption

Weather conditions affect the two groups differently (Table 1). Rain reduces member ridership more than casual (−252 **vs** −129), suggesting members are more exposed to precipitation disrupting fixed commuting routes. Snow is selected only for members (−226, Prob = 1.00) and not for casual riders, who appear to opt out entirely on extreme weather days rather than sustain a partial reduction.
Season effects are not selected in any BSTS model (Prob ≈ 0 across all four models). Once temperature and weather category are controlled for, calendar season provides no additional explanatory power.

5.4 BSTS vs OLS

Coefficient directions are consistent across both methods, confirming the core findings are robust to model choice. BSTS substantially outperforms OLS in R^2 (Table 2) because its local linear trend absorbs slow-moving temporal variation that OLS misattributes to calendar dummies. This is why season is significant in OLS (`seasonFall`: +443, $p < 0.001$) but not selected in any BSTS model.

	OLS	BSTS Baseline	BSTS Interaction
Member	0.72	0.932	0.933
Casual	0.76	0.864	0.870

Table 2: In-sample R^2 across all models. BSTS substantially outperforms OLS; adding interaction terms yields negligible further improvement.

6 Discussion and Recommendations

Three findings drive three recommendations:
1. Workday status is the primary demand differentiator. Members commute; casual riders do not. Weekday capacity and station allocation should be optimized for predictable member commuting patterns, which are stable regardless of weather.
2. Weather forecasts matter for casual demand, but mainly on weekends. The interaction model shows that on workdays, temperature has a weaker effect on casual demand (fewer casual trips happen regardless of conditions). Promotional campaigns targeting casual riders should focus on non-working days with favorable forecasts.
3. Season is a proxy, not a cause. Planning should be driven by weather forecasts, not calendar season. Rainy-day incentives (e.g., discounted casual trips) could help retain casual riders during adverse weather and build habitual usage.

6.1 Limitations

The analysis is city-level and does not capture station-level heterogeneity. The data cover one city across three years. MCMC convergence was monitored visually but not formally diagnosed with $\hat{R}$ statistics. Future work could apply a difference-in-differences design to evaluate whether targeted interventions shift the member-to-casual ratio as predicted.

7 Conclusion

Member and casual Citibike riders respond to the same conditions in opposite directions: workdays boost member demand and suppress casual demand by a combined 1,000 rides per day. Temperature affects both groups similarly, and season effects disappear once weather is controlled for. The practical implication is straightforward: these are two distinct products that require separate demand management strategies.

References

[1] Scott, S.L., & Varian, H.R. (2014). Predicting the present with Bayesian structural time series. *International Journal of Mathematical Modelling and Numerical Optimisation*, 5(1/2), 4–23.

[2] Citibike NYC. (2024). System Data. https://citibikenyc.com/system-data

[3] Open-Meteo. (2024). Historical Weather API. https://open-meteo.com